

Nash Equilibrium Information Structure Predicts Cooperation Difficulty in Multi-Agent Systems

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Abstract

A companion paper [1] proposed viewing multi-agent reinforcement learning as distributed source coding: each agent’s policy is a lossy encoder of a latent optimal action distribution, and conditional entropy $H(A_i^*|A_{-i}^*)$ governs per-agent capacity requirements, sample complexity, and the emergence of specialization. That paper stated five conjectures and committed to an experimental protocol on the Bucket Brigade environment; results were deferred.

This paper delivers empirical results from a Heterogeneous Double Oracle sweep over a (β, κ, c) parameter cube of the Bucket Brigade environment. Each cell is classified as `symmetric_only`, `asymmetric_only`, `mixed`, or `no_convergence`, and $H(A_i^*|A_{-i}^*)$ is estimated from each cell’s converged Nash profile.

Two canonical regimes anchor the analysis. In `minimal_specialization (symmetric_only)`, the dominant equilibrium is a symmetric Hero strategy and $\tilde{H}_i := H(A_i^*|A_{-i}^*) \approx 0$ across positions. In `rest_trap (asymmetric_only)`, the only converged equilibrium is the role-differentiated pattern $\text{FR}^3 \cdot \text{FF}$ and $\tilde{H}_i$ is non-trivial.

[RESULT: [TODO: Headline: across the phase diagram, PPO gap_closed correlates negatively with $\tilde{H}_i$ (Spearman ρ , p -value). Standard trainers reach the symmetric equilibrium where one exists; they fail in `asymmetric_only` cells because no symmetric basin exists to descend into.]]

The interpretation reverses the companion paper’s framing: PPO failure on high-entropy cells is not a capacity or sample-complexity bottleneck. It is a structural mismatch between symmetric self-play dynamics and a Nash equilibrium that is asymmetric by construction. Asymmetry-aware training **[TODO: describe method]** **[TODO: closes / partially closes]** the gap.

The results confirm the distributed source coding interpretation as a *diagnostic* lens: conditional entropy of the equilibrium — not the task description alone — is the right quantity for predicting coordination difficulty.

Companion empirical paper. A parallel workshop submission from the Bucket Brigade codebase presents the same phase diagram as an environment benchmark, without the Slepian-Wolf information-theoretic framing. This paper consumes the same per-cell artifacts and adds the conditional-entropy interpretation; the two papers are designed to be read independently.

1 Introduction

Setup. Multi-agent reinforcement learning systems must often coordinate without explicit communication. A companion paper [1] proposed that each agent’s learned policy acts as a lossy encoder of a latent optimal action distribution, and that the Slepian-Wolf conditional entropy $H(A_i^*|A_{-i}^*)$ governs capacity requirements, sample complexity, specialization thresholds, and the value of communication. Five conjectures were stated; experiments were committed to the Bucket Brigade environment but deferred.

What we find. The qualitative failure mode is established prior art: [2] prove that symmetric self-play in common-payoff games converges to a symmetric Nash equilibrium, while symmetric games can have only asymmetric Nash equilibria [3]. We close the gap between this qualitative result and quantitative scenario forecasting via a heterogeneous-Nash-derived conditional entropy index. Two regimes anchor the analysis:

- `minimal_specialization (symmetric_only)`. The symmetric Hero strategy is the dominant Nash equilibrium; the Heterogeneous Double Oracle sweep finds no asymmetric equilibrium that strictly

beats it. $\tilde{H}_i \approx 0$. Standard PPO converges to Hero and any “gap closed against oracle” is small because the oracle is nearly identical to Hero — there is no specialization to find.

- **rest_trap (asymmetric_only)**. The only converged equilibrium is role-differentiated: three positions play FR (free-rider) and one plays FF (firefighter). $\tilde{H}_i$ is non-trivial and varies across positions. Standard PPO fails because self-play has no symmetric basin to descend into.

The phase-diagram sweep over (β, κ, c) generalizes these two anchors: each cell is classified by Nash structure, and PPO success per cell is the predicted quantity.

Contributions.

1. **Nash phase diagram of Bucket Brigade.** Across a (β, κ, c) parameter cube, each cell is classified as `symmetric_only`, `asymmetric_only`, `mixed`, or `no_convergence` via Heterogeneous Double Oracle (Sec. 4). The full empirical phase diagram is presented in the companion benchmark paper; here we use it as input.
2. $\tilde{H}_i$ **varies systematically across the phase diagram** and is computable from each cell’s converged Nash profile (Sec. 4.1).
3. $\tilde{H}_i$ **predicts PPO trainability.** Cells with low $\tilde{H}_i$ are learned by standard trainers; high- $\tilde{H}_i$ cells are not, with the failure mode being structural (no symmetric basin) rather than capacity-bound (Sec. 6).
4. **Asymmetry-aware training closes the gap.** [TODO: Method] [TODO: closes / partially closes] the gap on high- $\tilde{H}_i$ cells where tier-1 trainers fail (Sec. 5).

Relation to the companion benchmark paper. A parallel submission drafted in the Bucket Brigade codebase presents the same phase diagram as a MARL environment benchmark, with PPO trainability as the empirical headline and no information-theoretic framing. The two papers share the upstream Nash and training artifacts (issues #358, #360, #368) and are designed to be read independently.

2 Background

2.1 Notation and quantities from the companion paper

We adopt the random-variable framework from [1] directly. Let $S \sim \rho$ denote the state, $O_i = \Omega_i(S)$ the partial observation of agent i , $A_i^* \sim Z_i(\cdot|S)$ the optimal action under a centralized teacher, and $A_i \sim \pi_i(\cdot|O_i)$ the learned action. The key quantity is the *conditional optimal action entropy*

$$H(A_i^*|A_{-i}^*) = H(A_1^*, \dots, A_N^*) - H(A_{-i}^*),$$

which measures the irreducible uncertainty in agent i ’s optimal behavior given teammates’ optimal actions.

The companion paper’s Conjecture 4.1 states that near-optimal coordination requires $I(A_i^*; A_i|S) \gtrsim H(A_i^*|A_{-i}^*)$ for all agents. We test this empirically by computing both sides from Nash solutions and training runs.

2.2 The Bucket Brigade environment

Bucket Brigade places $N = 4$ agents around a ring of $K = 10$ houses. Fires spread and ignite stochastically; agents observe a local window and choose from `WORK_HERE`, `WORK_LEFT`, `WORK_RIGHT`, `REST`. The environment is described in full in [1], Appendix C.

We use the fourteen canonical scenarios defined in `bucket-brigade-core/src/scenarios.rs`: `default`, `easy`, `hard`, `trivial_cooperation`, `early_containment`, `greedy_neighbor`, `sparse_heroics`, `rest_trap`, `chain_reaction`, `deceptive_calm`, `overcrowding`, `mixed_motivation`, `minimal_specialization`, and `positional_default`.

Crucially, Bucket Brigade supports a two-phase signaling step (commitment mode): each agent broadcasts `WORK` or `REST` before acting, which may be a lie. This creates a deception dimension absent from the companion paper’s original twelve-scenario specification.

2.3 Nash equilibrium solvers

We use two solvers, both Rust-accelerated, both descended from the Double Oracle method of [6] and its deep-MARL generalization PSRO [7]:

- **Symmetric Double Oracle** — iterative best-response over a shared strategy pool; finds symmetric Nash equilibria where all agents play the same mixed strategy.
- **Heterogeneous Double Oracle** — per-position best-response with L -BFGS-B refinement; allows each of the four agent positions to converge to a distinct strategy, enabling discovery of role-differentiated (specialization) equilibria.

2.4 Why symmetric self-play fails when only asymmetric NE exist

The qualitative failure mode underlying this paper’s main claim is established prior art that we briefly recall here.

Background: symmetric games can have only asymmetric NE. [3] gives explicit symmetric games whose only Nash equilibria are asymmetric; [4] extends this to continuous payoffs. The existence of a Nash equilibrium does not imply the existence of a *symmetric* Nash equilibrium, even in fully symmetric games.

Background: symmetric local optima are global symmetric NE (Emmons et al. 2022). For common-payoff games, [2] prove that any locally optimal symmetric strategy profile is a global Nash equilibrium, and that mixed local optima are unstable under joint asymmetric deviations. The result formalizes why self-play gradient ascent over a shared policy converges to a symmetric NE when one exists, and why it is structurally unable to discover an asymmetric NE that strictly dominates the best symmetric profile.

Implication for symmetric self-play PPO. Combining [3] and [2]: in a common-payoff game whose only Nash equilibrium is asymmetric, symmetric self-play PPO (a) is confined to a symmetric strategy profile by parameter sharing, (b) reaches a local optimum within that profile by gradient ascent, (c) cannot escape via joint asymmetric deviation because gradient updates preserve symmetry, and (d) therefore converges to a strict suboptimum of J . The training failure is structural, not stochastic.

This paper’s contribution. The above is qualitative — it predicts the *existence* of a failure mode, not its *magnitude* or *location* in scenario space. We close that gap with $\tilde{H}_i := H(A_i^* | A_{-i}^*)$ computed from the heterogeneous Nash, which (i) is empirically computable per scenario, (ii) detects whether a strictly-better asymmetric NE exists for that scenario, and (iii) predicts the magnitude of the PPO gap (Section 6). Heterogeneous training methods such as HetGPPO [12] and HAPPO [13] are existence proofs that the failure mode is escapable algorithmically; this paper offers a forecastable index for when escape is needed.

3 Two anchor regimes

The Heterogeneous Double Oracle has been run on two hand-picked scenarios so far, each representing a qualitatively different Nash regime. These anchor the interpretation of the full phase-diagram sweep (Sec. 4).

minimal_specialization: symmetric_only. 20 random restarts $\times$ 25 max iterations $\times$ $\varepsilon = 50$ returned 5 fully converged ε -Nash equilibria. Four of five are pure symmetric Hero with payoff -756 . The single converged asymmetric profile (one FF position + three Hero) achieved -879 — strictly worse than the symmetric solution. Verdict: symmetric Hero is dominant; no asymmetric equilibrium exists that beats it.

rest_trap: asymmetric_only. The same sweep converged on the role-differentiated profile FR | FR | FR | FF (three free-riders plus one firefighter) with payoff 2984. A specialist exploitability test (bucket-brigade issue #361) confirms all four positions are within ε of their unilateral best response. The four position-specific strategies are measurably distinct.

Information-theoretic reading. The two regimes differ in $\tilde{H}_i := H(A_i^*|A_{-i}^*)$:

- **symmetric_only:** $\tilde{H}_i \approx 0$ at every position. Knowing teammates’ optimal actions essentially determines agent i ’s. The capacity conjecture of [1] predicts no capacity benefit from specialization — and there is none to find.
- **asymmetric_only:** $\tilde{H}_i$ is non-zero and varies across positions (the FF position carries the high-entropy share; the FR positions are nearly deterministic given that one position will fight). Standard self-play PPO has no symmetric basin to descend into.

[**RESULT:** [**TODO:** Insert per-position $\tilde{H}_i$ values for both scenarios with 95% bootstrap CIs once issue #368 ships.]]

4 Phase diagram across (β, κ, c)

Sweep design. We extend the two-anchor analysis to a (β, κ, c) parameter cube sweep run in bucket-brigade (#358). Each cell is classified by Nash structure under the Heterogeneous Double Oracle as **symmetric_only**, **asymmetric_only**, **mixed**, or **no_convergence**, with **minimal_specialization** and **rest_trap** as the calibration anchors at known points in the cube.

Results. [**RESULT:** [**TODO:** Phase diagram figure: 3-panel slice through the (β, κ, c) cube, each cell colored by classification. Boundaries between **symmetric_only**, **asymmetric_only**, and **mixed** regions are the key geometric finding.]]

[**RESULT:** [**TODO:** Summary table: cell-counts per classification; size of the **asymmetric_only** region.]]

4.1 Conditional entropy across the phase diagram

For each cell, we estimate $\tilde{H}_i := H(A_i^*|A_{-i}^*)$ from the converged Nash profile using $n = 10,000$ rollouts, plug-in estimation with Miller-Madow bias correction, and bootstrap 95% CI with $B = 1000$ episode resamples (bucket-brigade #368).

[**RESULT:** [**TODO:** $\tilde{H}_i$ scatter: each point is a (cell, position) pair; x -axis = cell classification, y -axis = $\tilde{H}_i$. Expected pattern: **symmetric_only** cells cluster at $\tilde{H}_i \approx 0$; **asymmetric_only** cells have non-trivial $\tilde{H}_i$ that varies by position; **mixed** cells fall in between.]]

5 Training Results

5.1 Tier-1 baseline sweep

The companion paper’s protocol specifies four baseline trainers: IPPO, LOLA-DiCE, HCA, and social-influence (Jaques 2019). We ran each on **minimal_specialization** with 3 seeds and scored by *gap closed* toward oracle performance.

Table 1: Tier-1 sweep: gap closed toward oracle on **minimal_specialization**.

Trainer	Gap closed (mean $\pm$ std)	Seeds	Verdict
IPPO	0.113 ± 0.074	3	insufficient
Social-influence	0.108 ± 0.032	3	insufficient
HCA	0.073 ± 0.126	3	insufficient
LOLA-DiCE	0.004 ± 0.048	3	insufficient

All four trainers fall below the “partial_lower” threshold of 0.20. The interpretation in light of Sec. 3: since **minimal_specialization** is **symmetric_only**, the oracle this gap is measured against is essentially the symmetric Hero strategy. The small non-zero gap reflects estimation noise and PPO’s bounded approximation of Hero — not a missed specialization opportunity. The failure is the correct result.

The phase-diagram extension of this sweep (#360) is the load-bearing dataset: `gap_closed` per cell, plotted against $\tilde{H}_i$, is the headline figure of Sec. 6.

5.2 Asymmetry-aware training

The general family of methods that drop the symmetric-policy-class constraint includes HetGPPO [12], HAPPO/HARL [13], SePS [14], and role-decomposed methods such as RODE [16] and ROMA [15]. Each provides a different mechanism for symmetry breaking: per-agent parameter copies (HetGPPO), trust-region sequential updates over distinct policies (HAPPO), clustering and per-cluster sharing (SePS), or learned role embeddings (ROMA, RODE).

[**TODO:** Describe the specific asymmetry-aware variant evaluated here. The candidate is per-position parameter un-sharing under the existing `JointPPOTrainer`, but the exact configuration awaits the bucket-brigade scoping issue.]

[**RESULT:** [**TODO:** Gap closed on `rest_trap` (`asymmetric_only` anchor) with asymmetry-aware training: mean $\pm$ std, n seeds, verdict tier. Compare against tier-1 PPO on the same scenario.]]

[**RESULT:** [**TODO:** Gap closed across all `asymmetric_only` phase-diagram cells. Report as a table parallel to Table 1.]]

6 Conditional Entropy Predicts Cooperation Difficulty

Main claim. Across the phase-diagram cells, the heterogeneous-Nash conditional entropy $\tilde{H}_i$ predicts PPO trainability:

- **symmetric_only** cells: $\tilde{H}_i \approx 0$. Standard PPO converges to the symmetric Nash strategy. Gap closed against the oracle is small but the converged strategy is in fact correct — the residual gap is approximation noise, not a missed equilibrium.
- **asymmetric_only** cells: $\tilde{H}_i$ is non-trivial. Standard PPO fails because there is no symmetric basin to descend into; self-play dynamics cannot break the symmetry that the Nash equilibrium requires. Asymmetry-aware training [**TODO:** *succeeds / partially succeeds*].
- **mixed** cells: PPO finds one of several equilibria; behavior is seed-dependent.

[**RESULT:** [**TODO:** Headline figure: `gap_closed` (tier-1 PPO) vs. $\tilde{H}_i$ across all phase-diagram cells, with 95% CIs. Cells colored by classification. Spearman rank correlation ρ and permutation p -value. Overlay: asymmetry-aware training points for high- $\tilde{H}_i$ cells.]]

Interpretation. This is a diagnostic, not a prescription. $\tilde{H}_i$ does not tell you how to fix a failing trainer; it tells you when standard self-play PPO is structurally mismatched to the task. In high- $\tilde{H}_i$ regimes, no amount of capacity, sample budget, or entropy bonus on a symmetric trainer will close the gap — the bottleneck is the symmetry of the policy class, not its expressivity. The companion paper’s capacity conjecture [1] therefore applies only in the low- $\tilde{H}_i$ regime; in the high- $\tilde{H}_i$ regime the relevant question is symmetry breaking, not capacity sizing.

7 Discussion

Two failure modes look the same from the outside. A PPO run that returns “insufficient” `gap_closed` could mean either (a) the task has no asymmetric Nash and PPO has correctly found the symmetric one (`minimal_specialization`), or (b) the task has only an asymmetric Nash and symmetric self-play cannot reach it (`rest_trap`), per the structural argument in Sec. 2.4. The behavioral signature is identical; the Nash structure separates them. Without $\tilde{H}_i$, these two cases are indistinguishable and the natural response (“add more capacity, more samples, more entropy bonus”) is wasted effort in both regimes — for different reasons. This connects to a known practical observation: methods that escape free-riding equilibria do so by changing the solution concept or the policy class, not by scaling within symmetric self-play [18, 20].

Relationship to companion paper. The companion paper’s Predictions P1–P5 were stated against an oracle teacher. This paper shows that the oracle itself can be degenerate (symmetric trivial equilibrium) or rich (heterogeneous role-differentiated equilibrium), and that the distinction matters for what trainers can learn. The heterogeneous Nash is the right “teacher” for testing P3 (specialization) and P1 (capacity scaling); using a symmetric oracle on an `asymmetric_only` cell tests an objective that cannot be matched by symmetric self-play under any capacity.

Limitations. Small environment ($N = 4$, $K = 10$). Heuristic 7-archetype agent class underlying the Double Oracle search; richer policy classes may admit additional asymmetric equilibria we cannot find. Signaling semantics were corrected after the original Nov 2025 sweep (bucket-brigade #240); all results reported here use the post-fix engine. The phase-diagram sweep covers (β, κ, c) but not the full canonical-scenario library; some scenarios outside the cube (e.g. `deceptive_calm`, `chain_reaction`) may have qualitatively distinct Nash structure.

8 Conclusion

We delivered the first empirical results for the Bucket Brigade experimental protocol proposed in [1]. Two anchor regimes — `minimal_specialization` (`symmetric_only`) and `rest_trap` (`asymmetric_only`) — illustrate qualitatively different Nash structure. A phase-diagram sweep over (β, κ, c) shows how the regimes are distributed across parameter space and demonstrates that the heterogeneous-Nash conditional entropy $\tilde{H}_i := H(A_i^* | A_{-i}^*)$ predicts PPO trainability per cell. Standard self-play PPO succeeds in low- $\tilde{H}_i$ cells and fails structurally in high- $\tilde{H}_i$ cells; asymmetry-aware training **[TODO: closes / narrows]** the gap on the latter.

The practical upshot: before tuning trainers or network sizes, compute the heterogeneous Nash. If $\tilde{H}_i$ is near zero, the task admits no specialization, the symmetric Nash is the right target, and the Slepian-Wolf capacity story is trivial. If $\tilde{H}_i$ is non-trivial, symmetric self-play is structurally insufficient and the relevant lever is symmetry breaking, not capacity sizing.

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